

Block 4 – Session 2

Cloud Computing & Cloud Networking

Arquitectura de Xarxes

Universitat Pompeu Fabra

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 From VLANs to Overlay Networks
- 4 Cloud Network Architecture
- 5 Cloud Governance, Sovereignty & Compliance
- 6 Session Summary

Outline

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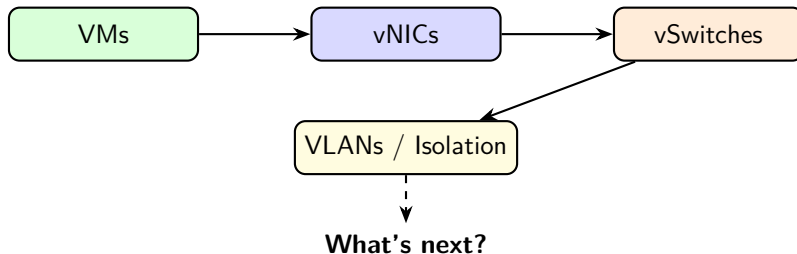
If We Can Virtualize Everything... Why Not Sell It?

*Virtualization is the technology.
Cloud computing is the business model.*

Learning objectives:

- 1 Understand cloud computing's definition and essential characteristics
- 2 Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) service models
- 3 Design a basic cloud network with VPCs (Virtual Private Clouds), subnets, and security controls

Recap – What We Virtualized (Session 1)



- Compute virtualization: VMs and containers
- Network virtualization: vNICs, vSwitches, vRouters, VXLAN overlays
- Together: the **building blocks** of the cloud

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	Cloud Computing
Run multiple VMs on one host	Offer compute as a service
Internal IT optimization	External/internal consumption
Manual provisioning possible	Automated, self-service
You own the hardware	Provider owns the hardware

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

Discussion: From Virtualization to Cloud

*Your company already runs VMs on its own servers.
Why would you move to the cloud instead of
keeping everything in-house?*

Hints: think about CapEx vs OpEx, scaling speed, and who manages what.

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Cloud Computing – Definition (NIST: National Institute of Standards and Technology)

*A model for enabling ubiquitous, convenient, **on-demand network access** to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.*

Source: NIST SP 800-145 (2011). Still the standard definition used industry-wide.

Five Essential Characteristics

Characteristic	What it means
On-demand self-service	Provision resources without human interaction
Broad network access	Available over the network from any device
Resource pooling	Shared infrastructure, multi-tenant
Rapid elasticity	Scale up/down automatically
Measured service	Pay only for what you use (metered)

- **All five** must be present to qualify as “cloud computing”
- If you need to call someone to get a VM → it is not cloud

Source: NIST SP 800-145, Mell & Grance, 2011.

On-Demand Self-Service

- Users provision resources through a **portal or API (Application Programming Interface)**
- No phone calls, no tickets, no waiting for approval
- Example: launch a VM in **30 seconds** via a web console

Analogy: vending machine (cloud) vs restaurant with a waiter (traditional IT).

Broad Network Access + Resource Pooling

Broad network access:

- Resources accessible from **any device** with network connectivity
- Standard protocols: HTTPS, SSH, REST APIs
- No proprietary client software needed

Resource pooling:

- Provider's resources serve **multiple tenants** simultaneously
- Tenants do not know (or care) where their resources are physically located
- Dynamically assigned based on demand
- Builds on the **multi-tenancy** concepts from Session 1

Rapid Elasticity + Measured Service

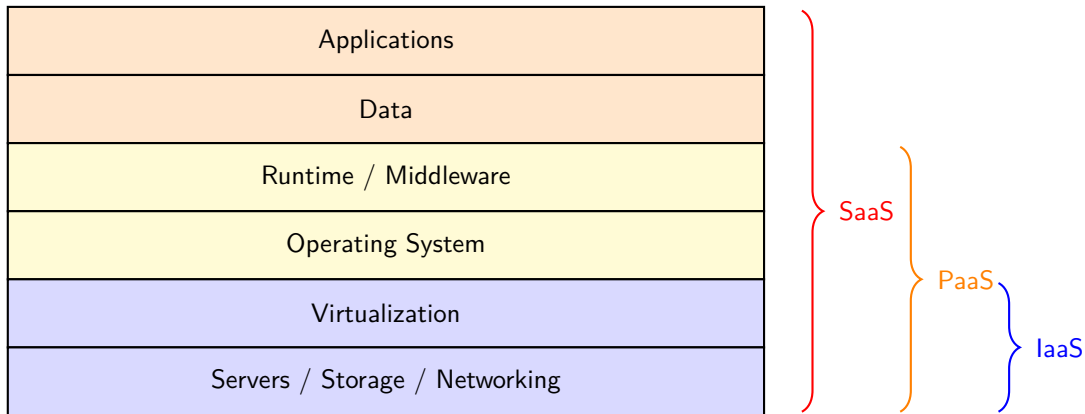
Rapid elasticity:

- Resources appear **unlimited** to the consumer
- Scale out (add) and scale in (remove) automatically
- Example: web store on Black Friday → 10× servers, then back to normal

Measured service:

- Usage is **metered** (CPU-hours, GB stored, GB transferred)
- **Pay-per-use** model → OpEx instead of CapEx
- Transparent monitoring and billing reports

Cloud Service Models – The Stack



IaaS – Infrastructure as a Service

- Provider gives you: **VMs, storage, networks**
- You manage: OS, middleware, applications, data
- **Maximum control, maximum responsibility**

Examples:

- **AWS EC2 (Elastic Compute Cloud):** virtual machines
- **Azure Virtual Machines**
- **Google Compute Engine**

Use case: “I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now.”

Source: AWS EC2 Documentation, docs.aws.amazon.com.

PaaS – Platform as a Service

- Provider gives you: **runtime environment + tools**
- You manage: only your **application code and data**
- No need to manage OS, patches, or scaling infrastructure

Examples:

- **Google App Engine**
- **AWS Elastic Beanstalk**
- **Heroku**

Use case: “I wrote a Python web app. Just run it, scale it, I do not care about servers.”

SaaS – Software as a Service

- Provider gives you: a **complete application**
- You manage: your **data and user configuration**
- No installation, no maintenance, no updates to worry about

Examples:

- **Google Workspace** (Gmail, Docs, Drive)
- **Microsoft 365** (Word, Excel, Teams)
- **Salesforce, Slack, Zoom**

Use case: “I need email for 500 employees. I do not want to run a mail server.”

Service Models – Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider

- IaaS → SaaS: **less control, less responsibility**
- Major providers (AWS, Azure, GCP) offer all three models

Discussion: Cloud Fundamentals

*A university wants to offer a web-based tool
for students to submit assignments.
Should they choose IaaS, PaaS, or SaaS? Why?*

Hints: consider the IT team size, maintenance burden, and how much control they need.

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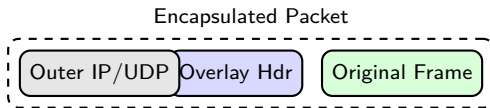
The VLAN Scalability Problem

- In Session 1 we used **VLANs** to isolate tenants on virtual networks
- VLANs use a **12-bit ID** → maximum **4,096** virtual networks
- Cloud providers host **millions** of tenants → VLANs are not enough
- Additional limitations:
 - VLANs are confined to a **single L2 domain**
 - Moving a VM to another host may require VLAN reconfiguration

We need a technology that scales beyond 4K networks and works across L3 boundaries.

Overlay Networks: Concept

- An **overlay network** is a virtual network built **on top of** the physical one
- Uses **encapsulation**: wrap virtual frames inside physical packets



- The physical network only sees the **outer headers**
- The virtual (tenant) traffic is hidden inside → **full isolation**

VXLAN: Overview

- **VXLAN** (Virtual eXtensible LAN): most widely used overlay protocol
- Encapsulates L2 frames in **UDP packets** over the physical network
- Uses a **24-bit VNI** (VXLAN Network Identifier)
 - $2^{24} = \mathbf{16 \text{ million}}$ virtual networks (vs 4,096 VLANs)
- Endpoints: **VTEPs** (VXLAN Tunnel Endpoints)
 - Encapsulate at source, decapsulate at destination

Source: IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.

VXLAN: How It Works



- VM A sends a frame → VTEP 1 wraps it in UDP with a VNI
- Travels over the physical network as a regular UDP packet
- VTEP 2 strips outer headers, delivers original frame to VM B
- VMs on the **same VNI** form an isolated L2 domain

Overlay Benefits for Scalability

- **16 million virtual networks** (vs 4,096 VLANs)
- Physical network does not need to know about tenants
- VMs can **migrate** across hosts without reconfiguring the underlay
- Decouples **virtual topology** from physical topology

This is the foundation of cloud networking. Next: how providers build **VPCs** on top of overlays.

Source: IETF RFC 8926, “Geneve: Generic Network Virtualization Encapsulation,” 2020.

Discussion: Overlay Networks

*Why can the physical network remain simple
if we use overlay networks?*

What are the trade-offs of encapsulation?

Hints: think about overhead (extra headers), MTU implications, and troubleshooting complexity.

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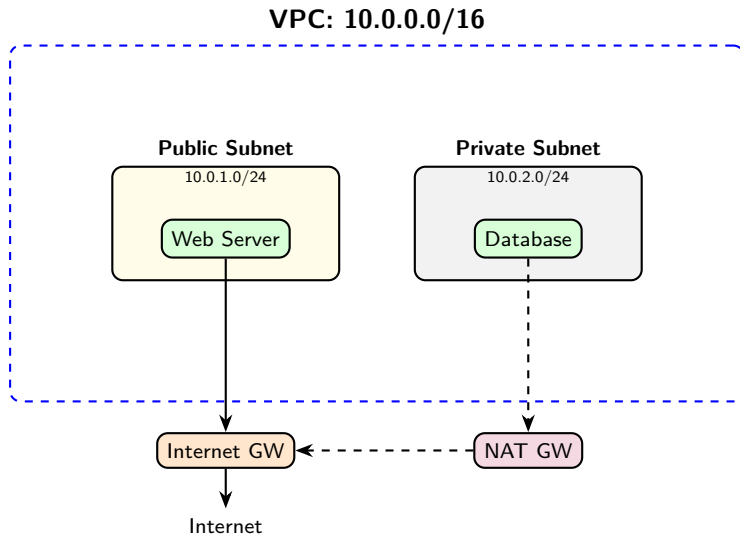
Virtual Private Cloud (VPC)

- A **VPC** is your own **isolated virtual network** inside a cloud provider
- Think of it as your private data center in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)
 - **Subnets** (subdivisions of the address space)
 - **Routing rules** and **security policies**

Key: the VPC is built on top of the **overlay network** concepts from Session 1.

Source: AWS, "Amazon VPC User Guide," docs.aws.amazon.com.

VPC Architecture



Subnets – Public vs Private

Public subnet:

- Has a route to an **Internet Gateway**
- Instances can have **public IP addresses**
- Accessible from the Internet (if security rules allow)
- Use for: web servers, load balancers, bastion hosts

Private subnet:

- **No direct route** to the Internet
- Instances only have **private IPs**
- Can access Internet outbound via **NAT Gateway**
- Use for: databases, internal services, application backends

Route Tables

- Each subnet is associated with a **route table**
- Route table = set of rules determining where traffic goes

Destination	Target	Subnet type
10.0.0.0/16	local (within VPC)	Both
0.0.0.0/0	Internet Gateway	Public
0.0.0.0/0	NAT Gateway	Private

- **Local route:** traffic within the VPC stays inside (always present)
- **Default route** (0.0.0.0/0): where to send everything else
- Public subnets → IGW; Private subnets → NAT GW

Internet Gateway vs NAT Gateway

	Internet Gateway	NAT Gateway
Direction	Bidirectional	Outbound only
Use case	Public-facing services	Private instances need updates
Assigned to	VPC (one per VPC)	Public subnet
Instance needs	Public IP	Only private IP

- Internet GW: allows the world to **reach** your public instances
- NAT GW: lets private instances **reach out** without being reachable
- Same **NAT concept from Block 3**, now as a managed cloud service

Security Groups – Instance-Level Firewall

- **Security group** = virtual firewall attached to an instance
- Rules specify allowed **inbound** and **outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Direction	Protocol	Port	Source/Dest
Inbound	TCP	80	0.0.0.0/0
Inbound	TCP	443	0.0.0.0/0
Inbound	TCP	22	10.0.0.0/16
Outbound	All	All	0.0.0.0/0

Source: AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

Network ACLs (Access Control Lists) – Subnet-Level Firewall

- **Network ACL (Access Control List)** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow both inbound AND outbound

Rule #	Direction	Protocol	Port	Action
100	Inbound	TCP	80	ALLOW
200	Inbound	TCP	443	ALLOW
*	Inbound	All	All	DENY

Source: AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

*You deploy a web application in a VPC.
The web server needs to be public, but the database
must not be reachable from the Internet. How?*

Hints: think about public vs private subnets, security groups, and NAT gateways.

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Cloud Governance

- **Governance** = policies and controls for managing cloud resources
- Key areas:
 - **Resource management**: who can create/delete what, tagging
 - **Cost control**: budgets, alerts, spending limits
 - **Access control**: Identity and Access Management (IAM)
 - **Compliance**: ensuring regulatory requirements are met

Without governance: resource sprawl, security gaps, surprise bills.

Data Sovereignty and GDPR

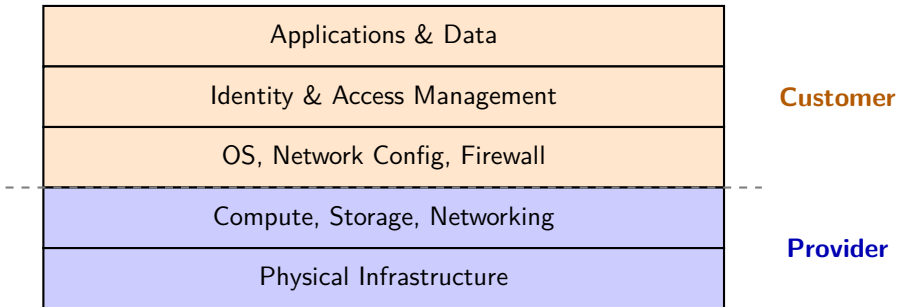
- **Data sovereignty**: data is subject to the laws where it is **stored**
- Cloud providers have data centers worldwide → your data location matters
- Risks: jurisdiction conflicts (EU data in US), government access (CLOUD Act)

GDPR (EU, 2018) – key principles:

- **Lawfulness + purpose limitation + data minimization**
- **Right to erasure**: users can request data deletion
- **Breach notification**: 72 hours to report
- **Relevance**: you must know **where** your data is and **who** can access it

Source: Regulation (EU) 2016/679, General Data Protection Regulation, 2018.

Shared Responsibility Model



- **Provider:** physical security, hardware, hypervisors, network fabric
- **Customer:** data, access policies, OS patches, application security
- The dividing line **shifts** depending on IaaS / PaaS / SaaS

Source: AWS, "Shared Responsibility Model," aws.amazon.com/compliance/shared-responsibility-model.

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Key Takeaways

- 1 Virtualization is the **technology**; cloud is the **business model**
- 2 Five NIST characteristics define true cloud computing
- 3 **IaaS / PaaS / SaaS** differ in who manages what
- 4 **VXLAN** overlays scale isolation to 16M networks (vs 4K VLANs)
- 5 A **VPC** is your isolated cloud network, built on top of overlays
- 6 **Security groups** (instance) + **ACLs** (subnet) = defense in depth
- 7 **Data sovereignty** and GDPR affect where and how you store data

Discussion

A Spanish startup stores user data in an AWS region in Ireland. A US law enforcement agency requests access. Who is responsible? What does GDPR say?

Think about: shared responsibility, data location, and which laws apply.

References

- ❶ NIST SP 800-145, Mell & Grance, "The NIST Definition of Cloud Computing," 2011.
- ❷ NIST SP 800-144, "Guidelines on Security and Privacy in Public Cloud Computing," 2011.
- ❸ AWS, "Amazon VPC User Guide," docs.aws.amazon.com.
- ❹ AWS, "Shared Responsibility Model," aws.amazon.com/compliance/shared-responsibility-model.
- ❺ Microsoft Azure, "Azure Virtual Network Documentation," docs.microsoft.com.
- ❻ Google Cloud, "VPC Documentation," cloud.google.com/vpc/docs.
- ❼ Regulation (EU) 2016/679, General Data Protection Regulation (GDPR), 2018.
- ❽ CSA, "Security Guidance for Critical Areas of Focus in Cloud Computing v5," 2022.
- ❾ Erl, Puttini & Mahmood, *Cloud Computing: Concepts, Technology & Architecture*, Prentice Hall, 2013.
- ❿ Kurose & Ross, *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- ⓫ IETF RFC 7348, "Virtual eXtensible Local Area Network (VXLAN)," 2014.
- ⓬ IETF RFC 8926, "Geneve: Generic Network Virtualization Encapsulation," 2020.